

National University of Computer and Emerging Sciences



Quiz 1

for

Object Oriented Programming Lab

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Section	BSE-2B
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Quiz 1

Quiz Duration: 35 minutes

Instructions:

- Read the questions carefully.
- Be careful with code indentation; improper indentation may result in negative marking.
- No tolerance for cheating or plagiarism.
- No extra time will be given. Failure to submit within the given time will result in zero marks.
- Submit .cpp file only.
- name the cpp file with your roll number. For example, 22L1234.zip.
- Submit your Quiz inside following folder:

[\\exam](#) > Fall 2025 Final Exam > Ahmad Jawad Mustasim > OOP > Quiz 1 > BSE-2B

Task

An area is represented as a grid containing lasers and safe cells.

- **1 represents a laser**
- **0 represents a safe cell**

A set of lasers is considered a trap if:

1. Each laser in the trap is aligned in different rows (no two lasers in the same row).
2. Lasers in the trap can share the same column.
3. Traps cannot “jump over” another row that contains a laser.

Write a C++ program that:

- Asks the user to enter the size of the area (number of rows and columns).
- Reads the values to fill the grid using only 1 for lasers and 0 for safe cells.
- Finds and displays the total number of laser traps Swiper has placed.

0	1	1	0	0	1
0	0	0	0	0	0
0	1	0	1	0	0
0	0	1	0	0	0

Trap Count = 8

Write a C++ program that:

1. Asks the user to enter:
 - Number of rows
 - Number of columns
2. Dynamically creates a **2D array using pointers**.
3. Takes input for the grid:
 - Only 0 and 1 are allowed.
 - Store values in the dynamic 2D array.
4. Finds and displays the **total number of traps** in the grid.
5. Properly deallocates all dynamic memory.

Restrictions

- Must use **2D dynamic arrays (pointer to pointer)**.
- Do not use vectors or built-in containers.
- Use loops and pointer logic only.
- Proper memory deallocation

Marking Scheme (Total = 10 Marks)

	Marks
Input validation	2
Dynamic Array creation	2
Trap Logic and Output	5
Proper memory deallocation	1